Academic Year: 2026-2027			
Program: Computer Engineering			
Year of Study	**Second Year**	**Semester**	**III**
Course Name	**Skill Based Lab - III (Web Technology II)**	**Course Code**	**CMLOR2VS201**
Name	**Tanmay Nilesh Shinde**	**Roll No**	**67**
Class & Div & Batch	**SYCM1-B**	**PRN No.**	**126BTCM2008**

Experiment No: 4	
Aim	Create a search bar that performs live filtering using AJAX
Learning Outcome	Implement AJAX to asynchronously fetch and display data from a REST API without reloading the webpage.

Program	<!DOCTYPE html>
	<html>
	
	<head>
	<title>Live Search</title>
	</head>
	
	<body>
	
	 Live User Search
	
	<input type="text" id="search" placeholder="Search User">
	
	<table border="1" cellpadding="10">
	<tr>
	<th>Name</th>
	<th>Email</th>
	<th>City</th>
	<th>Company</th>
	</tr>
	
	<tbody id="userTable"></tbody>
	</table>
	
	<script>
	let users = [];
	
	let xhr = new XMLHttpRequest();
	
	xhr.open("GET", "https://jsonplaceholder.typicode.com/users", true);

```
    xhr.onload = function () {
      if (xhr.status == 200) {
        users = JSON.parse(xhr.responseText);
        displayUsers(users);
      }
    };

    xhr.send();

    function displayUsers(data) {
      let table = document.getElementById("userTable");
      table.innerHTML = "";

      data.forEach(user => {
        table.innerHTML += `
          <tr>
            <td>${user.name}</td>
            <td>${user.email}</td>
            <td>${user.address.city}</td>
            <td>${user.company.name}</td>
          </tr>
        `;
      });
    }

    document.getElementById("search").addEventListener("input", function
() {
      let value = this.value.toLowerCase();

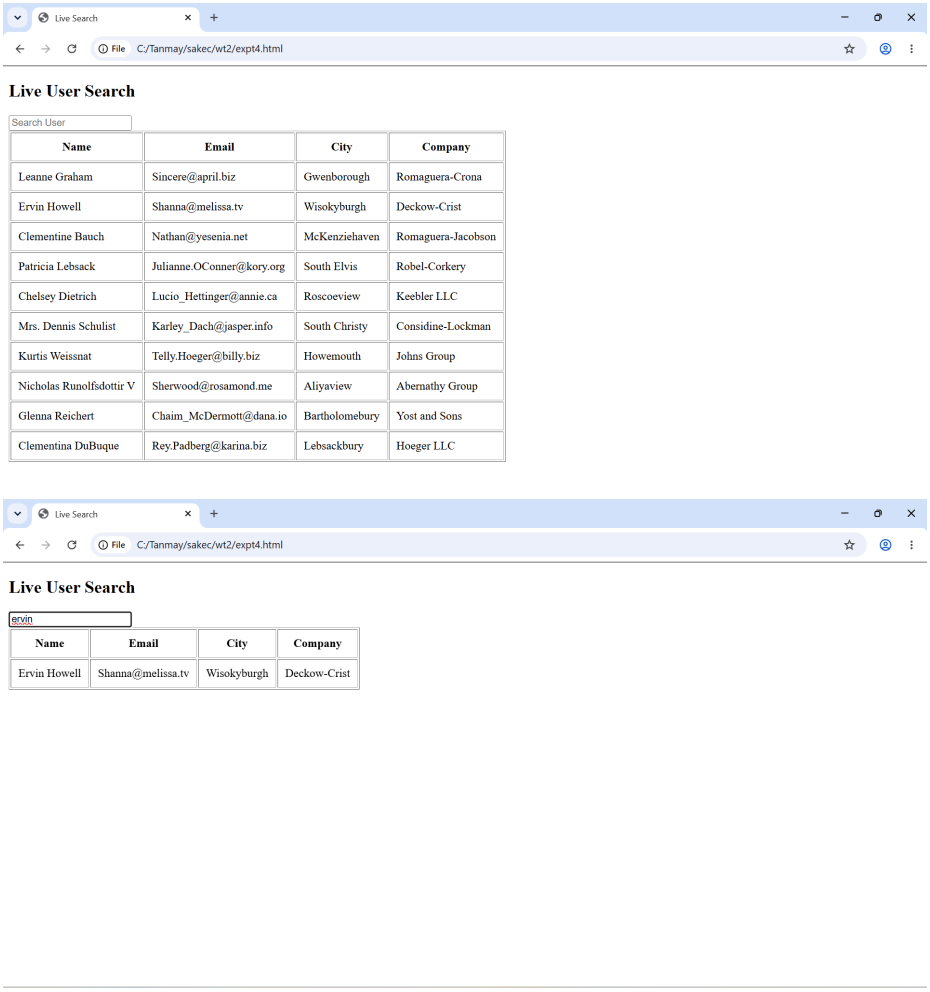
      let filtered = users.filter(user =>
        user.name.toLowerCase().includes(value) ||
        user.email.toLowerCase().includes(value) ||
        user.address.city.toLowerCase().includes(value) ||
        user.company.name.toLowerCase().includes(value)
      );

      displayUsers(filtered);
    });
  </script>

</body>

</html>
```

Output



Review Questions

Q1. Write a program to retrieve user details (Name, Email, City, and Company Name) from the following REST API using AJAX and display the data in a table.
REST API Endpoint: https://jsonplaceholder.typicode.com/users

Ans :

```
<!DOCTYPE html>
<html>
<head>
   <title>User Details</title>
</head>
<body>

   <h2>User Information</h2>

   <button onclick="loadUsers()">Load Users</button>

   <br><br>
```

```
<table border="1" cellpadding="10">
  <tr>
    <th>Name</th>
    <th>Email</th>
    <th>City</th>
    <th>Company Name</th>
  </tr>

  <tbody id="userTable"></tbody>
</table>

<script>
  function loadUsers() {

    let xhr = new XMLHttpRequest();

    xhr.open("GET", "https://jsonplaceholder.typicode.com/users", true);

    xhr.onload = function() {

      if (xhr.status == 200) {

        let users = JSON.parse(xhr.responseText);

        let table = document.getElementById("userTable");

        table.innerHTML = "";

        users.forEach(user => {

          table.innerHTML += `
            <tr>
              <td>${user.name}</td>
              <td>${user.email}</td>
              <td>${user.address.city}</td>
              <td>${user.company.name}</td>
            </tr>
          `;

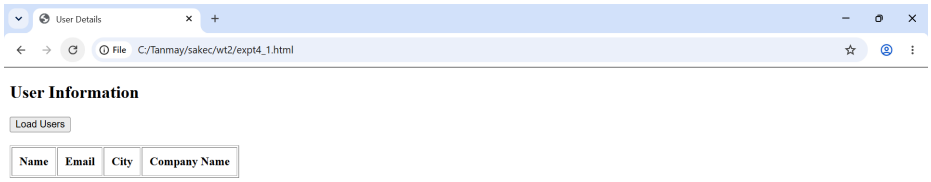
        });

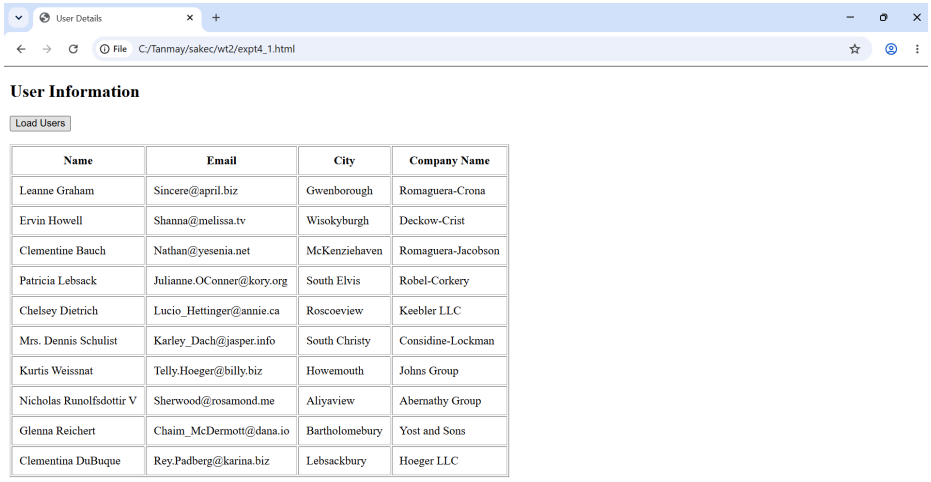
      } else {
        alert("Unable to load users");
      }
    };

    xhr.onerror = function() {
      alert("Error fetching data");
    };
```

```
        xhr.send();
    }
</script>

</body>
</html>
```





Name	Email	City	Company Name
Leanne Graham	Sincere@april.biz	Gwenborough	Romaguera-Crona
Ervin Howell	Shanna@melissa.tv	Wisokyburgh	Deckow-Crist
Clementine Bauch	Nathan@yesenia.net	McKenziehaven	Romaguera-Jacobson
Patricia Lebsack	Julianne.OConner@kory.org	South Elvis	Robel-Corkery
Chelsey Dietrich	Lucio_Hettinger@annie.ca	Roscoeview	Keebler LLC
Mrs. Dennis Schulist	Karley_Dach@jasper.info	South Christy	Considine-Lockman
Kurtis Weissnat	Telly.Hoeger@billy.biz	Howemouth	Johns Group
Nicholas Runolfsdottir V	Sherwood@rosamond.me	Aliyaview	Abernathy Group
Glenna Reichert	Chaim_McDermott@dana.io	Bartholomebury	Yost and Sons
Clementina DuBuque	Rey.Padberg@karina.biz	Lebsackbury	Hoeger LLC

Q2. Modify the above program to sort the retrieved user records alphabetically by City before displaying them on the webpage.

Ans :

```
<!DOCTYPE html>
<html>
<head>
   <title>Sort Users by City</title>
</head>
<body>
```

```
<h2>Users Sorted by City</h2>

<button onclick="loadUsers()">Load Users</button>

<br><br>

<table border="1" cellpadding="10">
  <tr>
    <th>Name</th>
    <th>Email</th>
    <th>City</th>
    <th>Company Name</th>
  </tr>

  <tbody id="userTable"></tbody>
</table>

<script>
  function loadUsers() {

    let xhr = new XMLHttpRequest();

    xhr.open("GET", "https://jsonplaceholder.typicode.com/users", true);

    xhr.onload = function() {

      if (xhr.status == 200) {

        let users = JSON.parse(xhr.responseText);

        users.sort((a, b) =>
          a.address.city.localeCompare(b.address.city)
        );

        let table = document.getElementById("userTable");

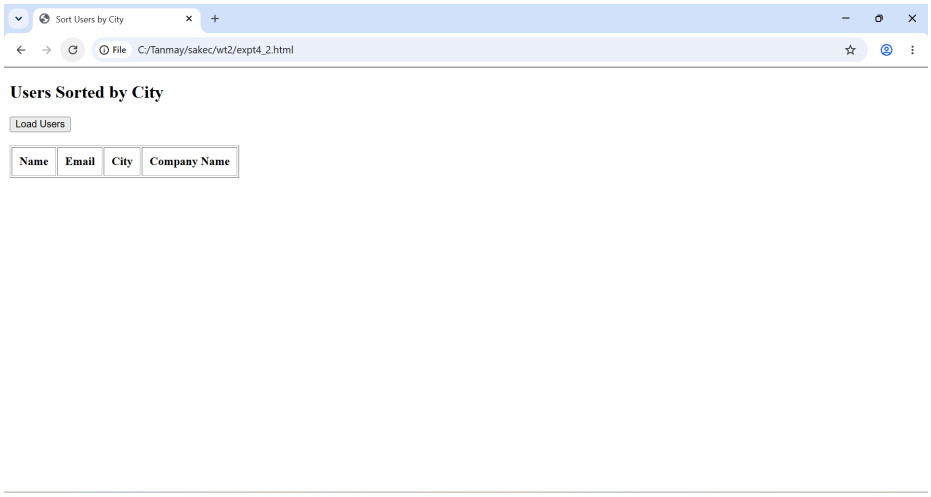
        table.innerHTML = "";

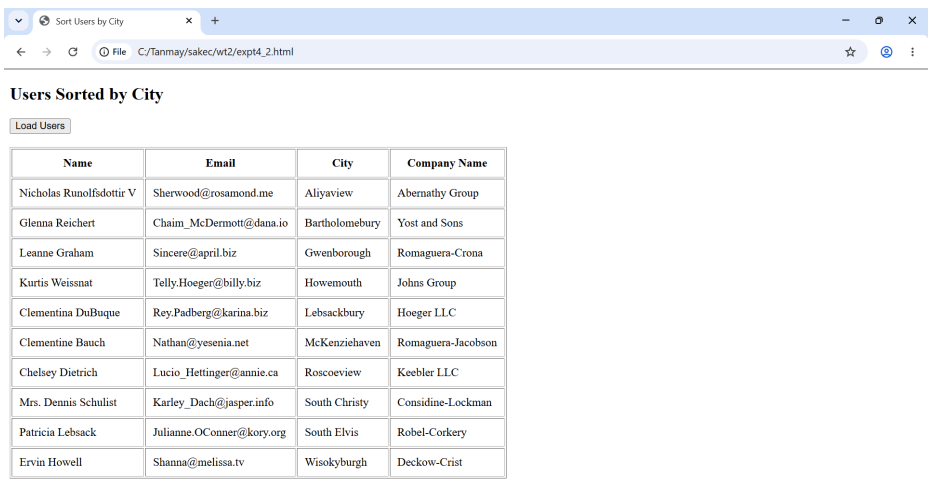
        users.forEach(user => {

          table.innerHTML += `
            <tr>
              <td>${user.name}</td>
              <td>${user.email}</td>
              <td>${user.address.city}</td>
              <td>${user.company.name}</td>
            </tr>
          `;

        });
```

```
            } else {
                alert("Unable to load users");
            }
        };
        xhr.onerror = function() {
            alert("Error fetching data");
        };
        xhr.send();
    }
  </script>
</body>
</html>
```





Name	Email	City	Company Name
Nicholas Runolfsdottir V	Sherwood@rosamond.me	Aliyaview	Abernathy Group
Glenna Reichert	Chaim_McDermott@dana.io	Bartholomebury	Yost and Sons
Leanne Graham	Sincere@april.biz	Gwenborough	Romaguera-Crona
Kurtis Weissnat	Telly.Hoeger@billy.biz	Howemouth	Johns Group
Clementina DuBuque	Rey.Padberg@karina.biz	Lebsackbury	Hoeger LLC
Clementine Bauch	Nathan@yesenia.net	McKenziehaven	Romaguera-Jacobson
Chelsey Dietrich	Lucio_Hettinger@annie.ca	Roscoeview	Keebler LLC
Mrs. Dennis Schulist	Karley_Dach@jasper.info	South Christy	Considine-Lockman
Patricia Lebsack	Julianne.OConner@kory.org	South Elvis	Robel-Corkery
Ervin Howell	Shanna@melissa.tv	Wisokyburgh	Deckow-Crist

Conclusion

Thus, we successfully created webpages using AJAX and XMLHttpRequest to retrieve user information from a REST API, implement live search filtering and sort user records alphabetically by city without reloading the webpage.